FILLFLOW AUTONOMOUS PLATFORM

FillFlow: Complete Demo Script & Gmail Architecture

Official Presentation Walkthrough, Feature Guide & Technical Pipeline

Platform: FillFlow AI Agent (Multi-Channel) **Engines:** Gmail API • Google Pub/Sub • Gemini AI **Format:** Executive Pitch & Technical Demo

1. Platform Feature Matrix

FillFlow converts corporate email inboxes and web channels into autonomous, 24/7 lead qualification and pipeline generation engines.

Module	Core Capabilities	Business Benefit
Gmail Automation	1-Click OAuth 2.0, Real-time Cloud Pub/Sub push webhooks, Gmail Watch API, RFC In-Reply-To thread tracking, fast memory deduplication.	Sub-second lead response time; eliminates polling lag and duplicate replies.
Gemini AI Intelligence	Multi-turn discovery dialogue, knowledge base grounding, intent classification (Inquiry, Follow-up, Pricing, Support), budget & timeline extraction.	Conversational replies that read like a seasoned human consultant.
AI Governance & Safety	3 Autonomy Modes (Autonomous, Semi-Autonomous, Safe Limited). Granular toggles for pricing, SLAs, refunds, and custom restricted keywords.	Eliminates hallucination risks & unauthorized financial commitments.
Project Brief Synthesis	Auto-generates structured briefs: technical stack, key risks, timeline brackets, and feature breakdown scored by complexity (Low/Med/High).	Reps enter sales calls with the architecture and requirements already defined.
CRM Pipeline & Quotas	Automatic Lead progression (`new` → `qualifying` → `qualified` → `brief_ready` → `converted`), 1-click Excel export, credit auto-locks.	Turnkey CRM workflow with strict operational budget controls.

2. Five-Minute Presentation Script (Acts 1 – 6)

Act 1: The Inbound Lead Bottleneck & Solution

0:00 – 0:45

Screen Action: Open FillFlow Dashboard (`/dashboard`). Highlight active lead counters, pipeline stages, and real-time activity metrics.

Voiceover: *"Every business loses high-value leads simply because incoming inquiries sit in an inbox for hours or days. Sales reps spend half their week manually qualifying leads, triaging spam, and typing the same repetitive answers.*

Meet **FillFlow** *— an autonomous AI agent built to live directly inside your existing business channels. It connects to your Google Workspace with a single click, detects incoming leads in real time, conducts intelligent multi-turn discovery conversations, and delivers fully structured project briefs before a human rep even opens the inbox."*

Act 2: The Gmail Connection & 1-Click Integration

0:45 – 1:45

Screen Action: Navigate to **Dashboard → Automations → Email**. Select the Google tab and click **"Connect Google Workspace / Gmail"**. Show the OAuth authorization modal and the instant 'Connected' badge with quota indicators.

Voiceover: *"Connecting your inbox takes less than 30 seconds. On the Email Automation tab, we click* **Connect with Google**.

We adhere strictly to the principle of least privilege: we only request permissions to read incoming client messages and send replies within authorized conversation threads.

The moment authorization completes, FillFlow registers a Google Cloud Pub/Sub webhook using the official **Gmail Watch API**. *That means zero polling delays — the exact millisecond a customer emails you, Google sends a secure push notification directly to our server."*

Act 3: Live Inbound Flow & Multi-Turn Intelligence

1:45 – 3:00

Screen Action: Split-screen view: Send a realistic client inquiry email to the connected company inbox (e.g., *"We need an on-demand logistics web platform, budget $20,000, target launch by Q3. What tech stack do you recommend?"*). Switch to FillFlow dashboard to show the lead created in real-time, then show the AI's professional reply in the sender's inbox.

Voiceover: *"Let's watch this live. A prospective client emails asking for an on-demand logistics platform with a $20,000 budget. Instantly, FillFlow's inbound engine validates the email, creates a new Lead profile, extracts budget and timeline, and consults the company's verified Knowledge Base.*

It then dispatches a context-rich reply addressing the technology recommendations and asking the next crucial qualification question. Notice how FillFlow preserves RFC `In-Reply-To` and `References` headers, keeping the entire email chain consolidated in one neat thread inside Gmail."

Act 4: AI Governance, Topic Guardrails & Approval Queue

3:00 – 4:00

Screen Action: Open **Settings → AI Control Center**. Show topic toggles (Services = Active, Custom Pricing = Restricted, SLA = Restricted). Show a message routed to the **Approval Queue** with editable response cards.

Voiceover: *"The number one concern enterprises have with AI is loss of control: 'What if the AI promises an unauthorized discount or misquotes contract terms?'*

FillFlow solves this with a granular **AI Permission Engine**. *You control autonomy topic by topic: General questions and capabilities are automated, while sensitive topics like pricing, custom discounts, and legal SLAs are automatically intercepted. When a lead asks for custom contractual terms, FillFlow either sends an approved holding response or drafts a proposed reply directly into the* **Approval Queue** *for your team to review, edit, and send with one click."*

Act 5 & 6: Automated Project Brief, CRM & Closing

4:00 – 5:00

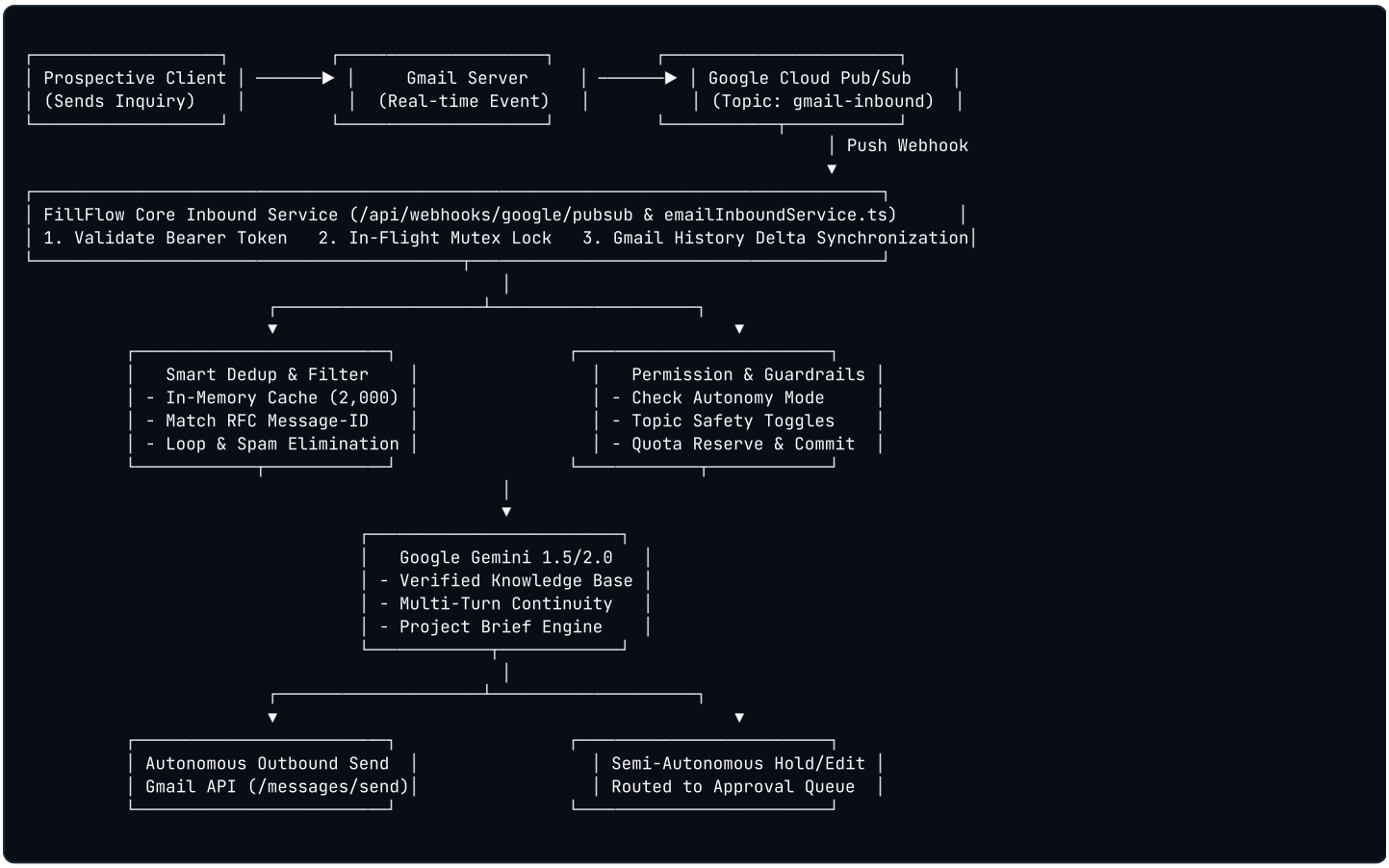
Screen Action: Click into the Lead Detail view. Display the synthesized **Project Brief** with feature complexity badges (Low/Medium/High) and click **"Export to Excel"**.

Voiceover: *"Once the lead completes discovery, FillFlow automatically synthesizes a full* **Project Brief**. *It categorizes each required feature by complexity, assesses architecture risks, estimates budget compatibility, and marks the lead as `brief_ready`.*

Your sales reps can download the entire pipeline into Excel and enter discovery calls 100% prepared. FillFlow converts your inbox into an automated, 24/7 revenue engine: zero missed inquiries, instant response times, and bulletproof safety controls."

3. Technical Gmail Connection Architecture

The entire email pipeline is event-driven, token-resilient, and fortified against duplicates and infinite loops.



4. Core Engineering Deep-Dive

1. Zero-Polling Real-Time Gmail Watch
Leverages Google Cloud Pub/Sub with the official `users.watch` API in `googlePubSubWebhookService.ts`. Eliminates polling delays, battery drain, and rate-limit hits.

2. Token Resiliency & Auto-Rotation
In `GoogleProvider.ts`, access tokens are checked on every request. If within 60 seconds of expiration, the provider transparently rotates using the offline refresh token.

3. Anti-Loop & Idempotency Armor
Dual-tier caching (in-memory LRU cache of 2,000 IDs + PostgreSQL snapshot comparison). Self-originating emails are immediately dropped to prevent infinite loops.

4. RFC Thread Preservation
Inbound Gmail `threadId` and RFC headers (`Message-ID`, `In-Reply-To`, `References`) are tracked end-to-end so outbound replies stay grouped in the client's inbox.

5. Quick Live CLI Verification

To demonstrate the end-to-end pipeline in a live technical evaluation, execute the test scripts below:

```
# Runs full multi-turn Gmail + Gemini discovery pipeline locally in seconds:
npx tsx scripts/test-real-gemini-live-e2e.ts

# Runs regression and safety guardrail verification:
npx tsx scripts/run-post-remediation-smoke-test.ts
```